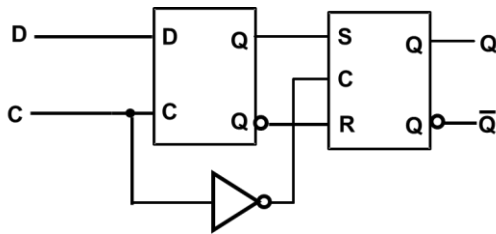


Digital Electronics Lab (CPE 462)
Lab 6
Sequential Circuit Design (Flip flops)

In this lab we will

- Design the schematic for the Master-Slave D Flip Flop and analyze its functionality
- Design the schematic for the Master-Slave D Flip Flop and analyze its functionality



PART 1: D-Master-Slave FF (Schematic)

Design the schematic for the D-Master-Slave FF and verify its functionality:

- 1- Measure the D-to-Q delay propagation delay of the FF
- 2- Measure the Clk-to-Q delay of the FF
- 3- Measure the setup time for the FF
- 4- Measure the hold time of the FF

PART 2: D-Master-Slave FF (layout)(Post lab)

Design the layout for the D-Master-Slave FF and verify its functionality:

- 1- Measure the D-to-Q delay propagation delay of the FF
- 2- Measure the Clk-to-Q delay of the FF
- 3- Measure the setup time for the FF
- 4- Measure the hold time of the FF